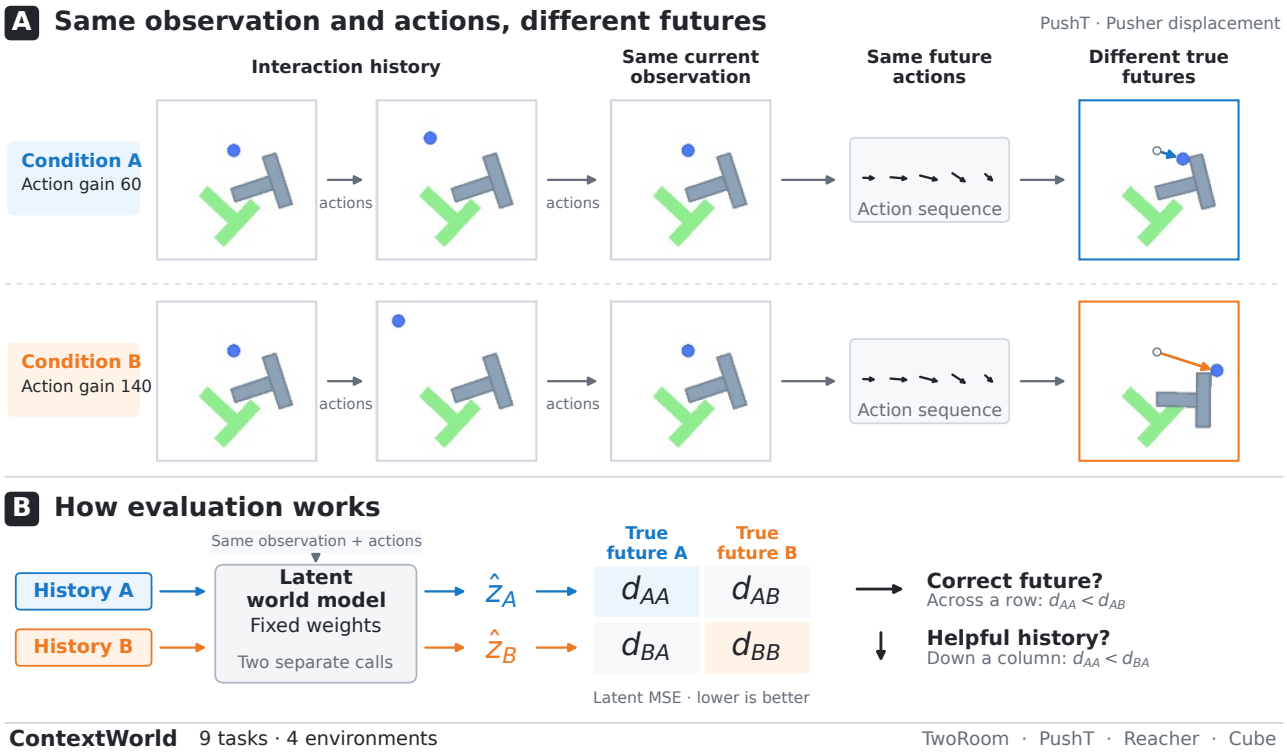


Figure 1. ContextWorld pairs interaction histories to test dynamics adaptation. A: Real PushT observations show two action gains. The current observation and future actions are identical, but the true futures differ. All images use the same crop; colored arrows mark the pusher's displacement from the shared current position. B: The same frozen model makes separate latent predictions. Matrix entries are MSEs between predicted and encoded true futures. Rows compare possible futures; columns compare histories for a fixed target. The displayed inequalities illustrate condition A; both conditions are scored.

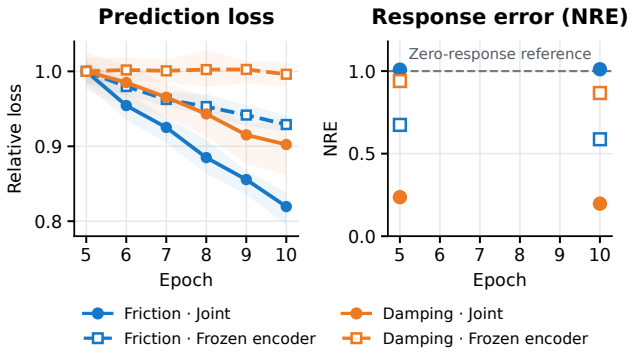


Figure 4. Prediction loss and conditional response. PLDM is trained on contact friction and motion damping, with the visual encoder jointly updated or frozen. Loss is normalized to epoch 5; bands show batch interquartile ranges. Training-query NRE is measured at epochs 5 and 10 only; 1 marks the zero-response reference. Within each task, both schemes start from the same Original checkpoint.

Prediction Loss and Adaptation

Weak evaluation performance may reflect poor generalization or conditional responses that were not learned well during training. We compare prediction loss with NRE on fixed training queries at epochs 5 and 10.

On friction, PLDM joint training reduces prediction loss by 18.05%, yet training-query NRE remains near the zero-response reference. On damping with the encoder frozen, loss approaches convergence while NRE improves but remains high. Changing the scheme helps: freezing yields a stronger conditional response on friction, whereas joint training performs better on damping.

These results expose a disconnect between prediction-loss optimization and history-conditioned adaptation, even on training data. Prediction loss aggregates errors across mixed data, whereas NRE directly measures error in the predicted differences between conditions.